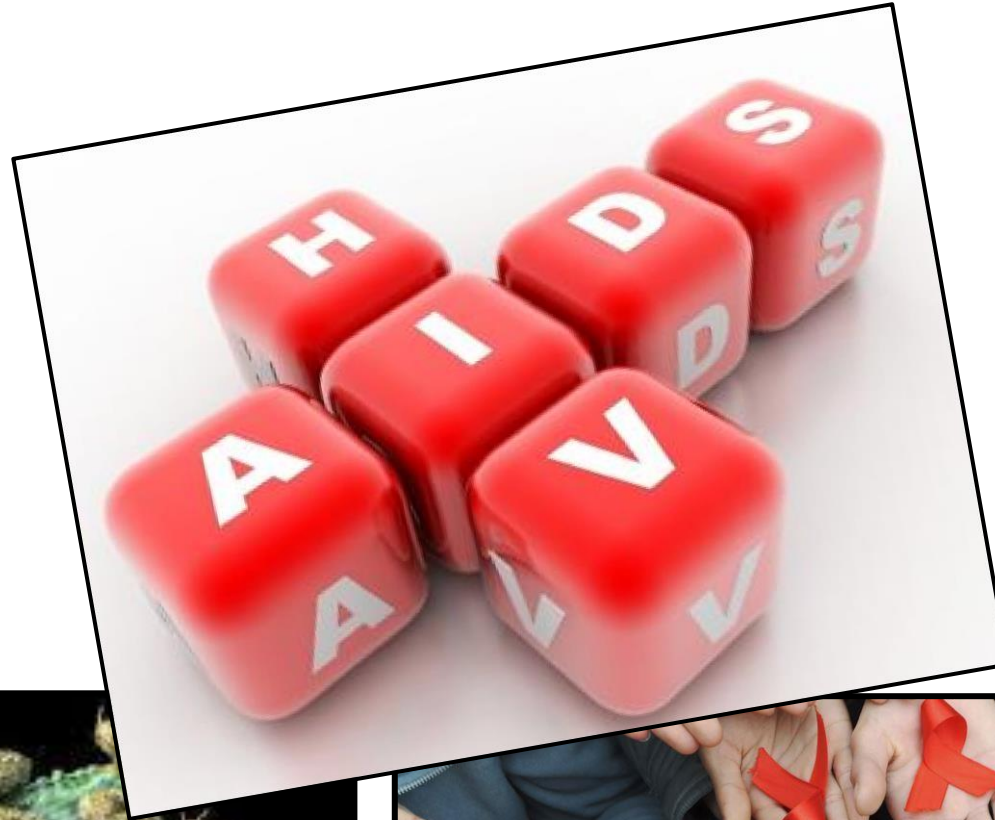


Antiretroviral Drugs



HIV Treatment considerations

- CD4 count (**concern when <200 cells/mm³**)
- HIV RNA copies – pretreatment levels
- Resistance
- Previous success/failure with antiretrovirals
- CD4+ T cell receptor – CXCR4 or CCR5?
- Pregnancy
- Drug interactions – CYP3A4
- Food effects – absorption consequences
- Additional infections – HBV, HCV, TB, etc
-



Co-infections common to HIV/AIDS

Opportunistic Pathogens:

- Tuberculosis (TB)
- Cryptococcal meningitis
- Cytomegalovirus
- Candidiasis
- Toxoplasmosis
- Cryptosporidiosis

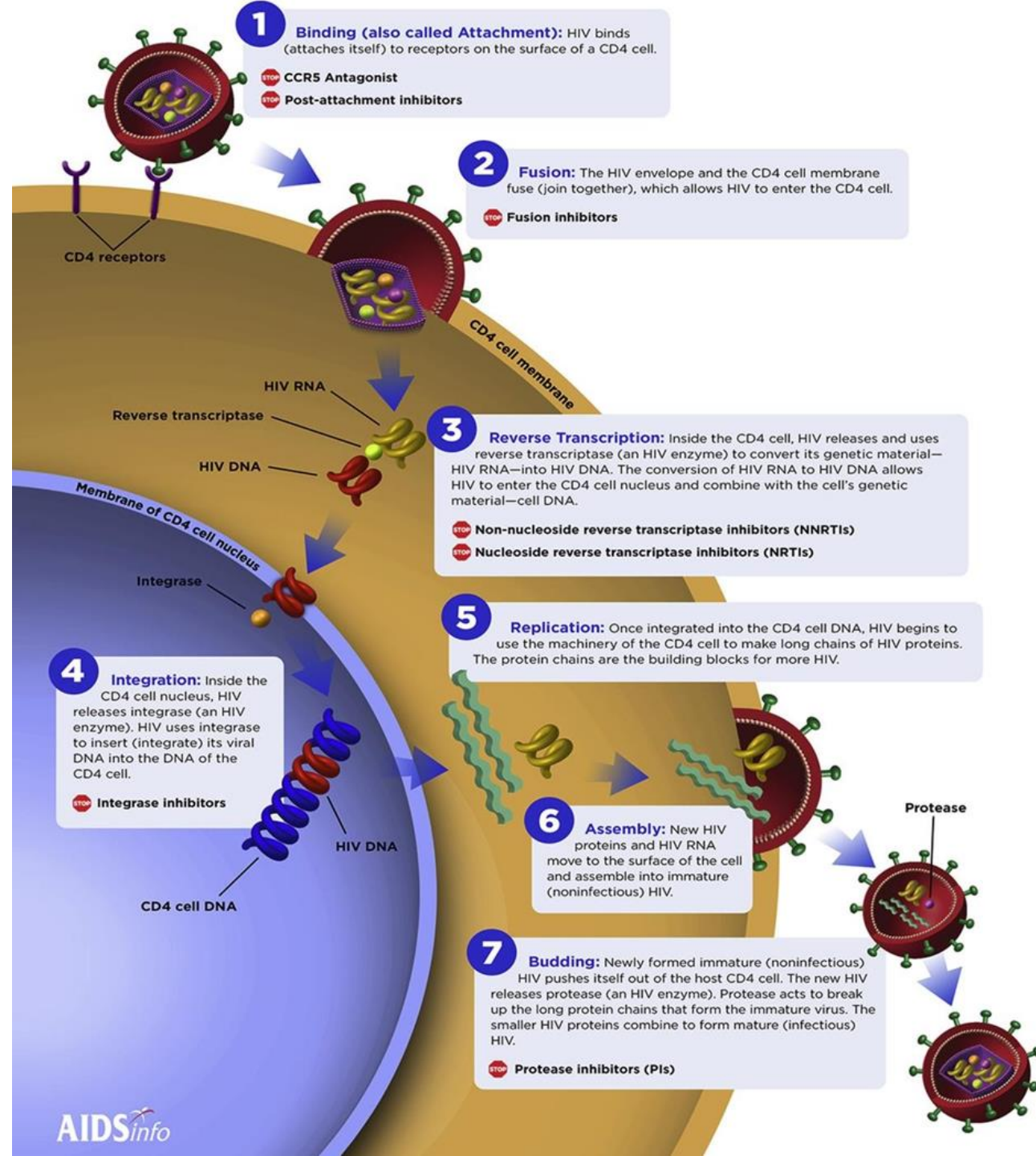
Cancers common to HIV/AIDS

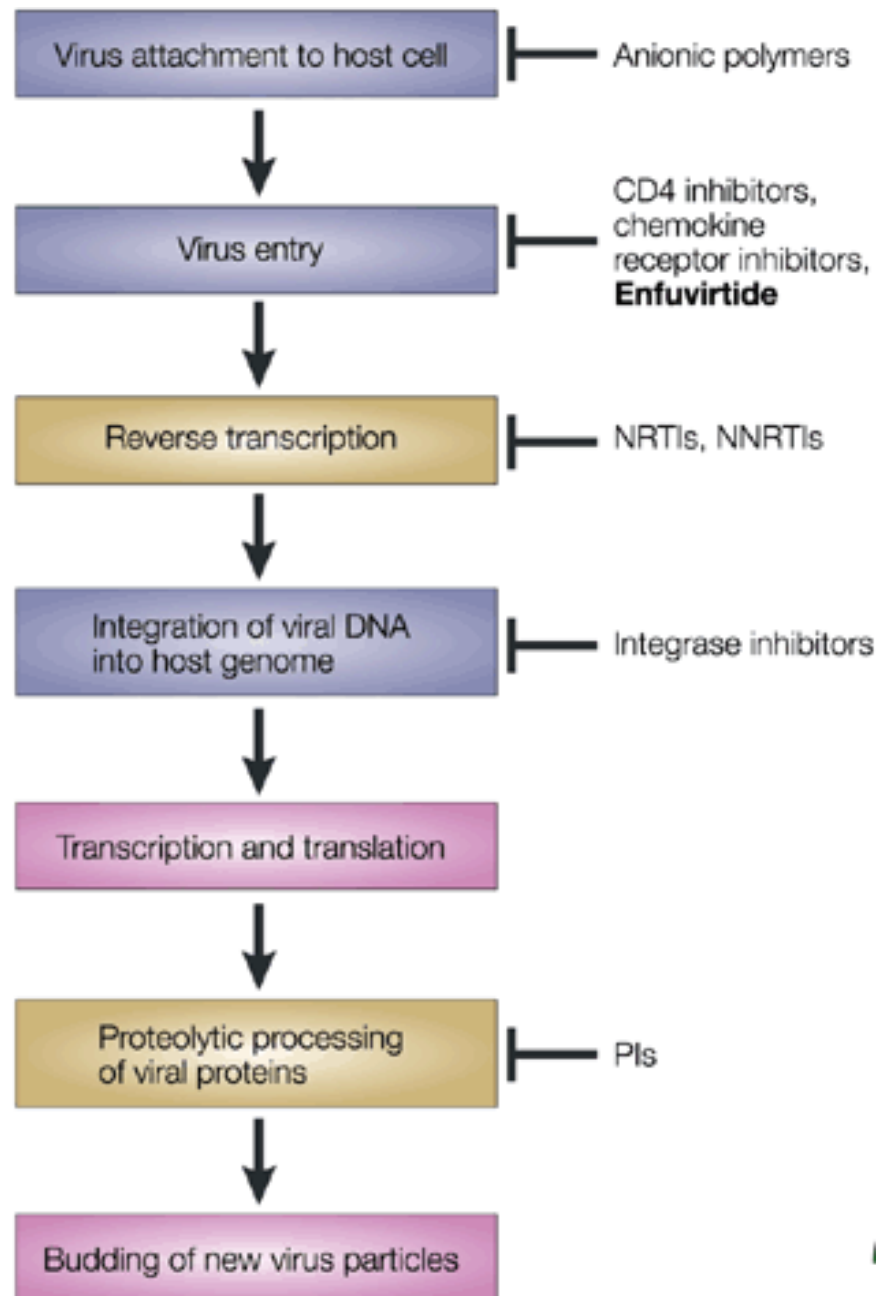
- Kaposi's sarcoma
- Lymphomas

Other complications

- Wasting syndrome
- Neurological complications
- Kidney disease

Cryptococcal meningitis	Tuberculosis	Severe bacterial infection
		
• 223,100 cases per year • 81% mortality rate • Represents 15% of AIDS-related deaths worldwide	• 1.2 million cases per year • 33% mortality rate • Represents 33% of AIDS-related deaths worldwide	• Exact statistics in HIV unknown • More than 6 million deaths per year among whole population; people with advanced HIV at higher risk





Treatment of HIV Infection: Mechanisms

Reverse Transcriptase Inhibitors

*nucleoside analogues (NRTI)

Emtricitabine (Emtriva)
Tenofovir (Viread)
Lamivudine (Epivir; 3TC)
Abacavir (Ziagen)

Non-Nucleotide Reverse Transcriptase Inhibitors (NNRTI)

Rilpivirine (Edurant)
Doravirine (Pifeltro)

Integrase Inhibitor

Dolutegravir (Tivicay)
Cabotegravir (Vocabria)
Bictegravir

Protease Inhibitors

Darunavir (Prezista)
Atazanavir (Reyataz)
*Cobicistat (Tyboost)
*Ritonavir (Norvir)

Fusion Inhibitors

Enfuvirtide (Fuzeon)
Maraviroc (Selzentry)
Fostemsavir (Rukobia)
Ibalizumab (Trogarzo)

Combinations decrease viral load, resistance and produce remission in some patients. Drugs do NOT eradicate the virus

Highly Active Antiretroviral Therapy (HAART)

An antiretroviral (ARV) regimen for a treatment-naïve patient generally consists of:

- two nucleoside reverse transcriptase inhibitors (NRTIs) administered in combination with:
- a third active ARV drug from one of three drug classes:
 - an integrase strand transfer inhibitor (INSTI),
 - a non-nucleoside reverse transcriptase inhibitor (NNRTI), or
 - a protease inhibitor (PI) + pharmacoenhancer

Pre-exposure Prophylaxis (PrEP)

- when people at very high risk for HIV take HIV medicines daily to lower their chances of getting infected. i.e., sexual or injection-drug-using partner who's HIV+

Post-exposure Prophylaxis (PEP)

- taking antiretroviral medicines (ART) after being potentially exposed to HIV to prevent becoming infected. Must be started within 72 hours after a possible exposure to HIV.



Nucleoside Reverse Transcriptase Inhibitors (NRTI)

Emtricitabine (Emtriva)

Tenofovir (Viread)

Lamivudine (Epivir; 3TC)

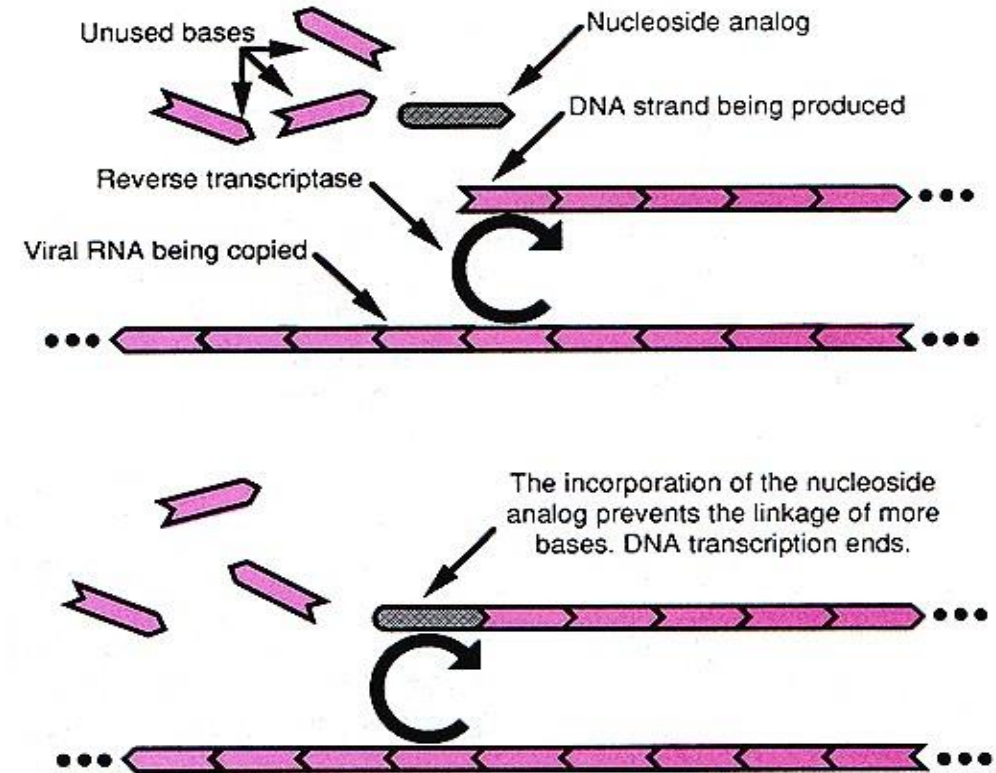
Abacavir (Ziagen)

} combination

} combination

Mechanism of action:

- Nucleoside analogue, requires phosphorylation (x3)
- incorporated into DNA, inhibits viral reverse transcriptase



Tenofovir (Viread) Emtricitabine (Emtriva)

- **Adenosine** (tenofovir) or **cytosine** (emtricitabine) analogues
 - Combined product - Truvada

Use:

- oral
- **First line of treatment for naïve HIV patients**
- Also used for PrEP
- *also monotherapy for HBV (tenofovir, inhibits HBV polymerase)*

Adverse effects

- GI, **flatulence**

Lamivudine (Epivir)

MOA - **Cytosine analogue**; Inhibits **HIV reverse transcriptase** and HBV polymerase

- oral
- Very effective, low toxicity – often used in treatment regimens
- Very well tolerated:
 - side effects include headache, fatigue, insomnia, GI
 - Safe in pregnancy

also Monotherapy for HBV

Abacavir (Ziagen)

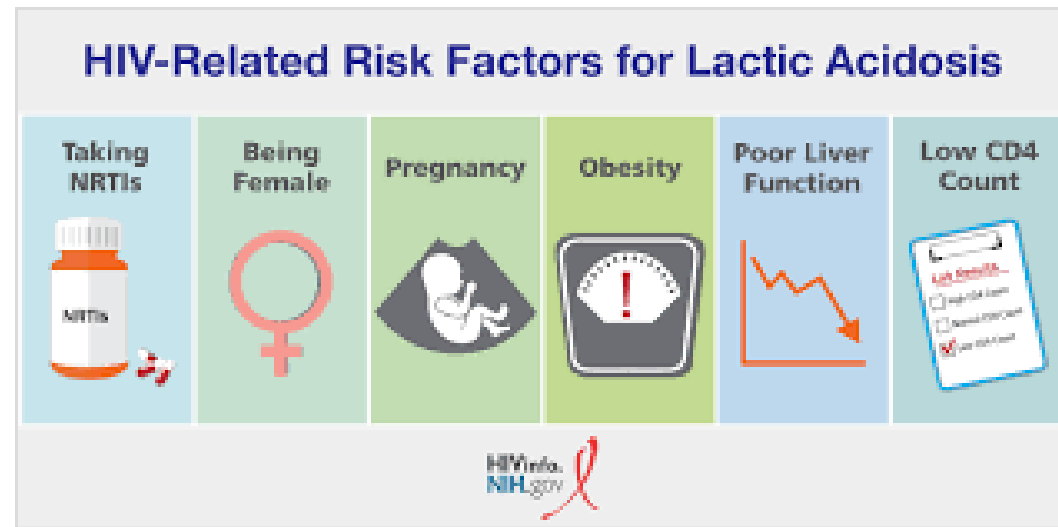
- **Guanosine** analogue
- Often in combination with lamivudine or zidovudine

Adverse effects:

- **Serious hypersensitivity** reactions occur in about 5% of patients; fever, malaise, skin rash, chills, sore throat and difficulty breathing
- **Screen for HLA-B*5701, only use in patient negative for HLA-B*5701**
- DISCONTINUE drug and DO NOT RESTART- ***restarting drug can be fatal!***
- nausea, vomiting, diarrhea

Serious Side Effects: All NRTIs

Lactic acidosis and **hepatotoxicity** are very rare, but serious side effects that may occur with any of **the nucleoside reverse transcriptase inhibitors** (more likely if the patient is obese, has previous liver disease, or takes the drugs for a long time).



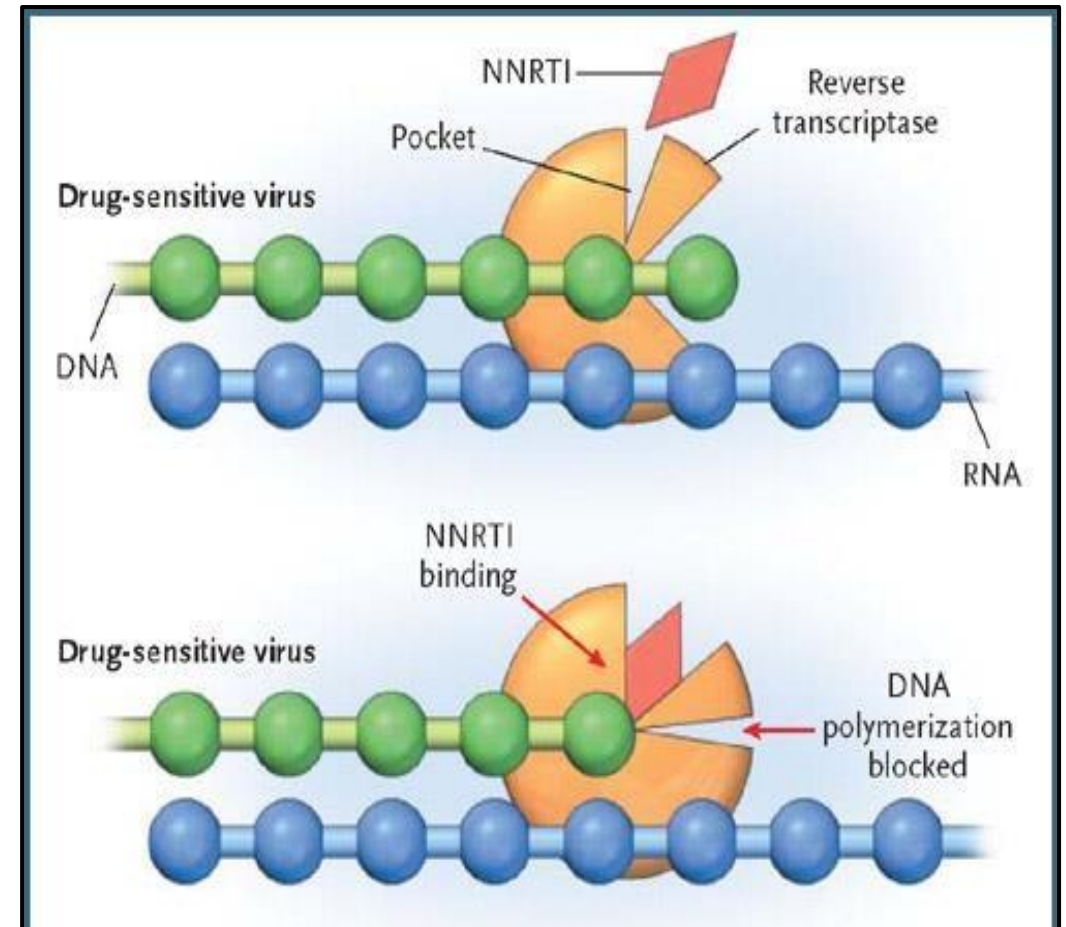
Non - Nucleotide Reverse Transcriptase Inhibitors (NNRTI)

Rilpivirine (Edurant)
Doravirine (Pifeltro)

MOA - bind directly to inhibit viral reverse transcriptase and prevent conversion of RNA to double strand DNA

Use – in combination with NRTIs, rapid resistance if used alone

*do not require phosphorylation for activity



Rilpivirine (Edurant)

- oral

Adverse effects:

- depression, increased cholesterol and headache
- Not recommended for patients with hepatitis co-infection, may increase liver enzymes

Doravirine (Pifeltro)

- oral

Adverse effects:

- CNS (headache, fatigue)
- Cardiovascular
- GI, rash

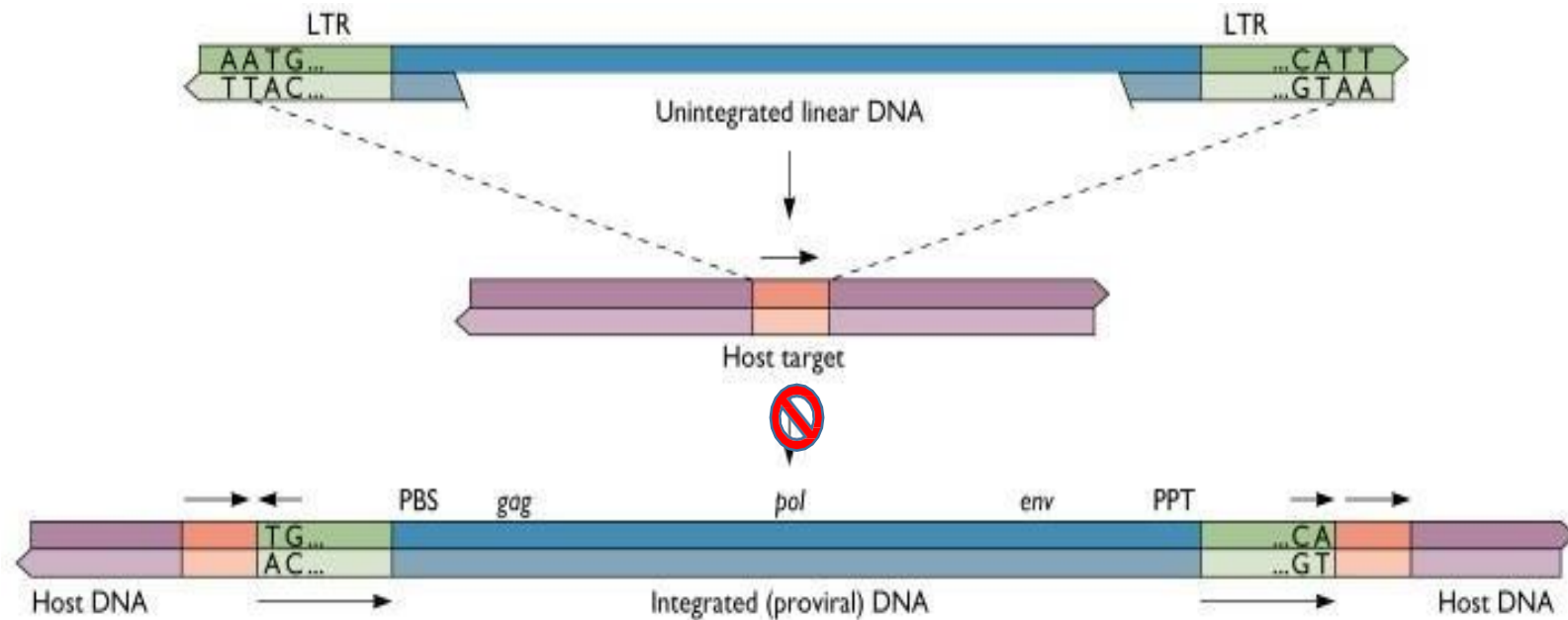
Integrase Inhibitors

Dolutegravir (Tivicay)
Cabotegravir (Vocabria)
Bictegravir

MOA - integrase inhibitor -

blocks the integrase enzyme
needed for replication,
inhibiting transfer of viral DNA
to host cell DNA

- typically used in combination with
NRTIs



Integrase Inhibitors

Dolutegravir (Tivicay)

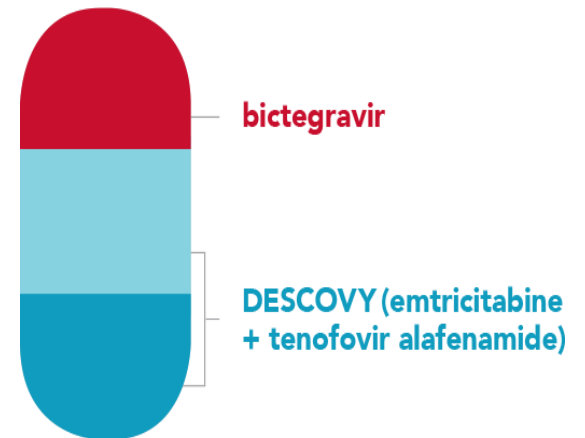
- Oral
- Adverse effects: headache, insomnia, GI, hyperglycemia, hepatitis (rare)

Cabotegravir (Vocabria)

- Oral, IM
- Adverse effects: rash, depression, headache, insomnia

Bictegravir

- Biktarvy: Complete treatment regimen – bictegravir, emtricitabine (NRTI), tenofovir (NRTI)
- Oral, taken once daily
- Adverse effects: nephrotoxicity, lactic acidosis, hepatotoxicity



Cabenuva FYI

- Approved January 2021
- Combination treatment with cabotegravir (Integrase inhibitor), rilpivirine (NNRTI)
- For individuals with undetectable virus
- ***First extended release injectable*** suspension administered 1x/month or 1x/2 months
- Also available orally, taken 1x/day for one month prior to administration of injectable.
- Adverse effects: hypersensitivity, hepatotoxicity, depression, injection site reaction



Protease Inhibitors

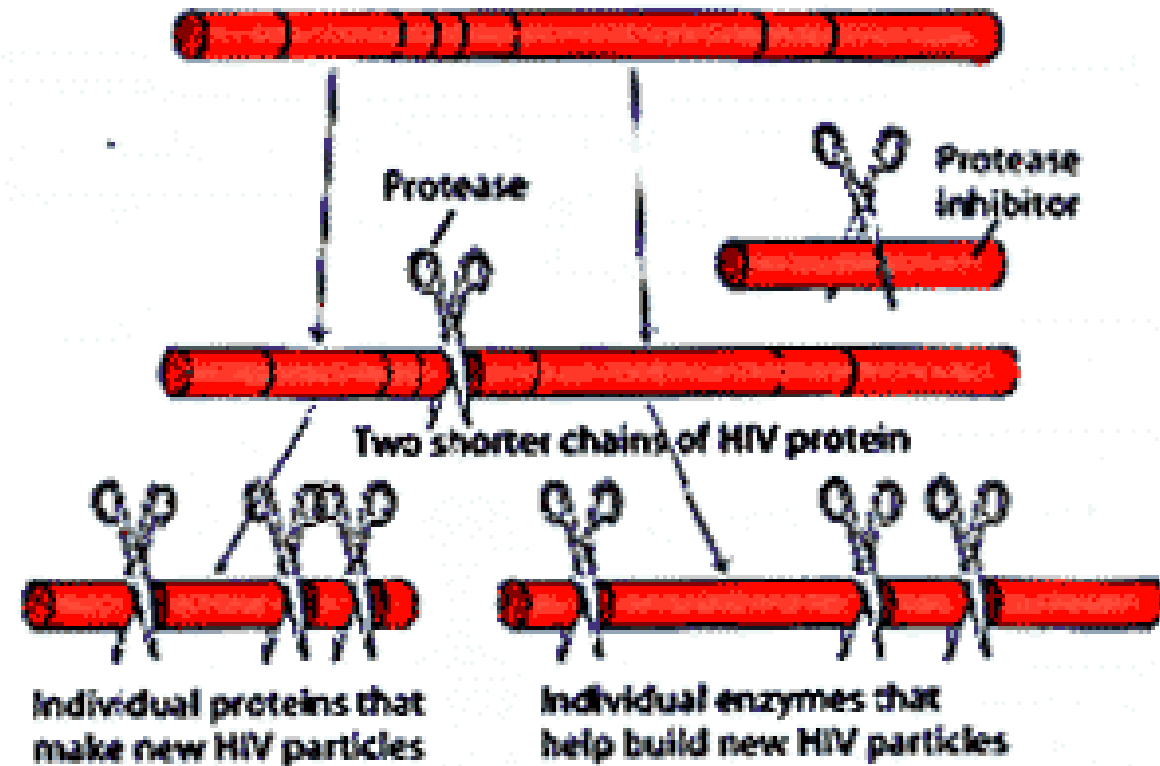
Darunavir (Prezista)

Atazanavir (Reyataz)

Ritonavir (Norvir)

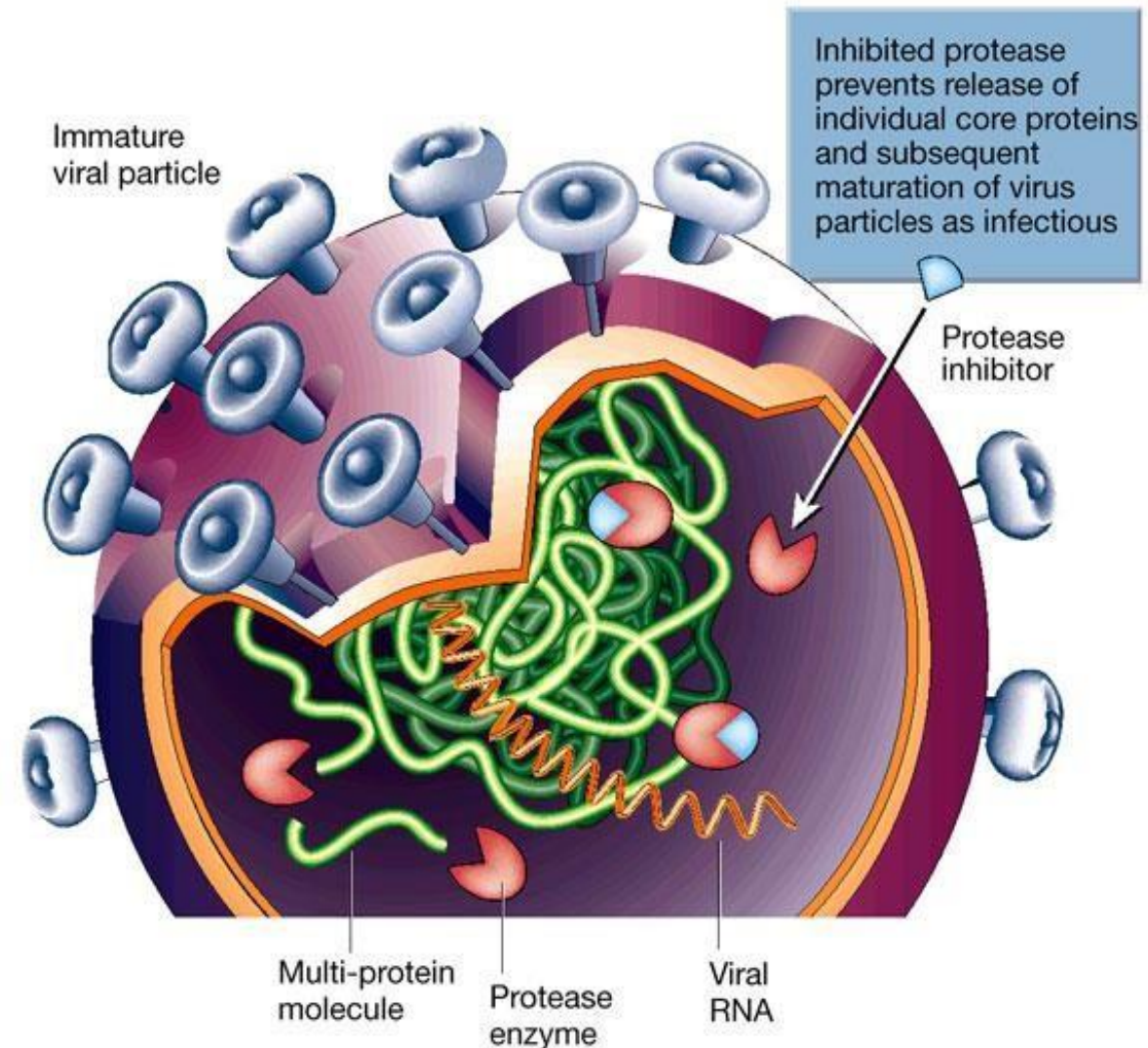
Cobicistat (Tybost)

❖ pharmacoenhancers



Protease Inhibitors

MOA - PI **bind to proteases** inhibiting their function: which is to digest long viral polypeptides into smaller, mature functional proteins. Viral particles are unable to mature and become infectious.



Protease Inhibitors - Common Pharmacokinetics

- oral
- Mutations and resistance common if used alone: **combine with NRTIs**
- PIs are metabolized by CYP3A4 – often given with ritonavir or **cobicistat**, which is a potent inhibitor of CYP3A4 to increase PI blood levels for lower dose/less frequent dosing.
- Many drug interactions:
 - **Drugs which induce CYP3A4 will decrease PI levels** (e.g. **rifampin**, phenytoin, phenobarbital, carbamazepine, ...).
 - St. John's Wort – do not use, increases metabolism of PIs!

KEY CHALLENGES OF HIV/TB CO-INFECTION

High disease burden

There are over 1.2 million cases of TB among people living with HIV.

TB therapy is difficult to take

Treatment lasts at least six months, which can affect adherence.

'Drug-to-drug interactions'

TB treatments can impact the effectiveness of HIV medicines.

Pharmacokinetic Enhancers

MOA

- Inhibits CYP3A4 - combined with other protease inhibitors to increase their bioavailability
 - “boost” or “pharmacoenhancer” - permits less frequent dosing and greater antiviral activity
- There will also be many drug interactions

Ritonavir (Norvir)

Adverse effects

- GI - vomiting, weakness and diarrhea
- burning, tingling, elevated liver enzymes, nausea
- Formulation contains **ethanol**; do not give with metronidazole, cephalosporins

Cobicistat (Tybost)

Adverse effects

- Hepatic, CNS (headache, dreams), rash

HIV Antiretroviral Pharmacokinetic Boosters **Ritonavir & Cobicistat**

MECHANISM
CYP3A4
INHIBITORS

↓ Drug Metabolism,
↑ Drug Levels

Boosted ARV's:

All Protease Inhibitors &
Elvitegravir (INSTI)

Ritonavir
(PI & Booster)
PI Side Effects & CYP
Drug Interactions

Cobicistat
(Booster Alone)
CYP Drug Interactions
but no PI Side Effects

Protease Inhibitors – Adverse effects



- **Altered body fat distribution-** buffalo hump and truncal obesity, with facial and peripheral atrophy
- **Insulin resistance** and hyperglycemia
- Increase in **serum cholesterol**; pharmacokinetic interaction with statins (pitavastatin and pravastatin OK)
- Spontaneous **bleeding** in patients with hemophilia A or B

Darunavir (Prezista)

- Now drug of first choice, when PI included in treatment regimen
- can be given with ritonavir

Adverse effects

- Safer and more effective than other protease inhibitors
- Rash, nausea and diarrhea most common side effects
- Headache, nightmares
- Contains a sulfonamide moiety – do not use in patients with sulfonamide allergy
- Must be taken with food – drug needs acid for absorption

Once-Daily PREZISTA® 800 mg with ritonavir 100 mg



Not shown at actual size with ritonavir 100 mg once daily.
Total number of pills taken will depend on overall regimen.

Atazanavir (Reyataz)

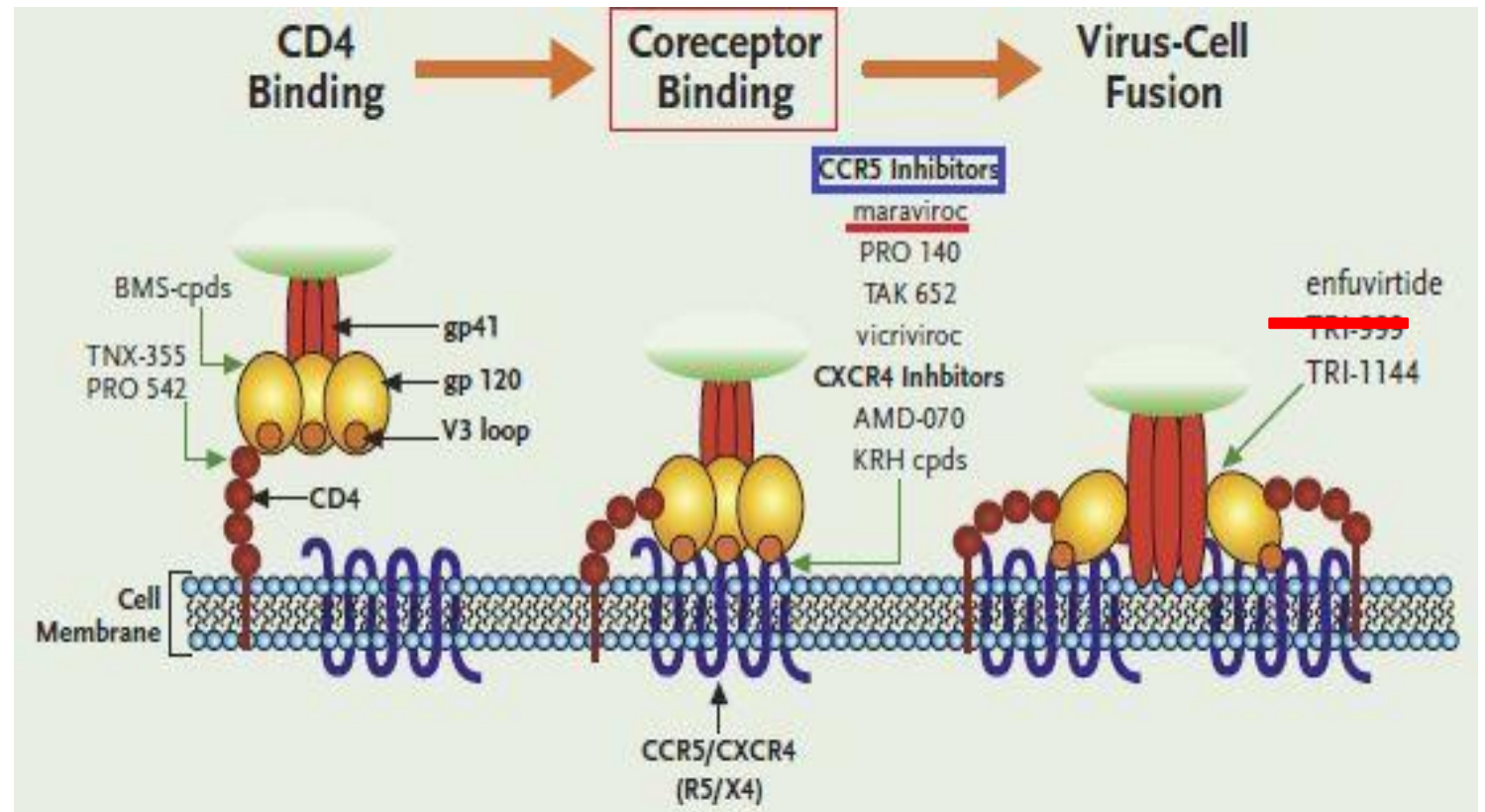
- **drug of second** choice; low incidence of side effects
- Once a day administration; taken orally with light meal to increase bioavailability
- Can be given with ritonavir

Adverse effects:

- **Less effect on body fat distribution** than other protease inhibitors
- May cause increases in **bilirubin** due to inhibition of UGT (uridinediphosphate-glucuronosyl transferase)1A1
- **Rash**, diarrhea, nausea is common, hyperglycemia may occur



Fusion Inhibitors



Enfuvirtide (Fuzeon)
Maraviroc (Selzentry)
Fostemsavir (Rukobia)
Ibalizumab (Trogarzo)

Enfuvirtide (Fuzeon)

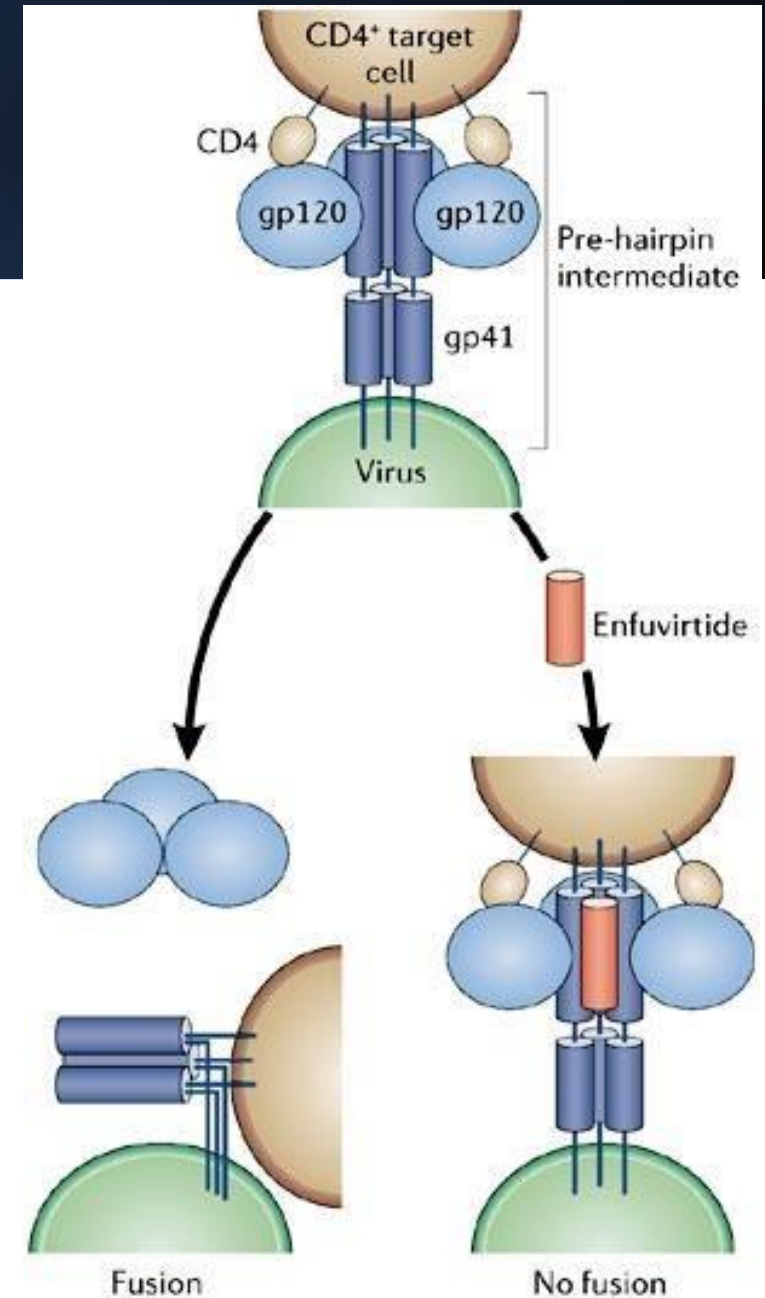
MOA - **fusion inhibitor** - binds to the **gp41** subunit of the viral envelope glycoprotein, prevents conformational change required for membrane fusion and viral entry

Use - patients with **advanced disease, treatment- experienced** and continue to show evidence of viral replication

- **subcutaneous injection** – inconvenient and expensive

Adverse effect - well tolerated

- injection site swelling, insomnia, headache
- increased likelihood of bacterial pneumonia



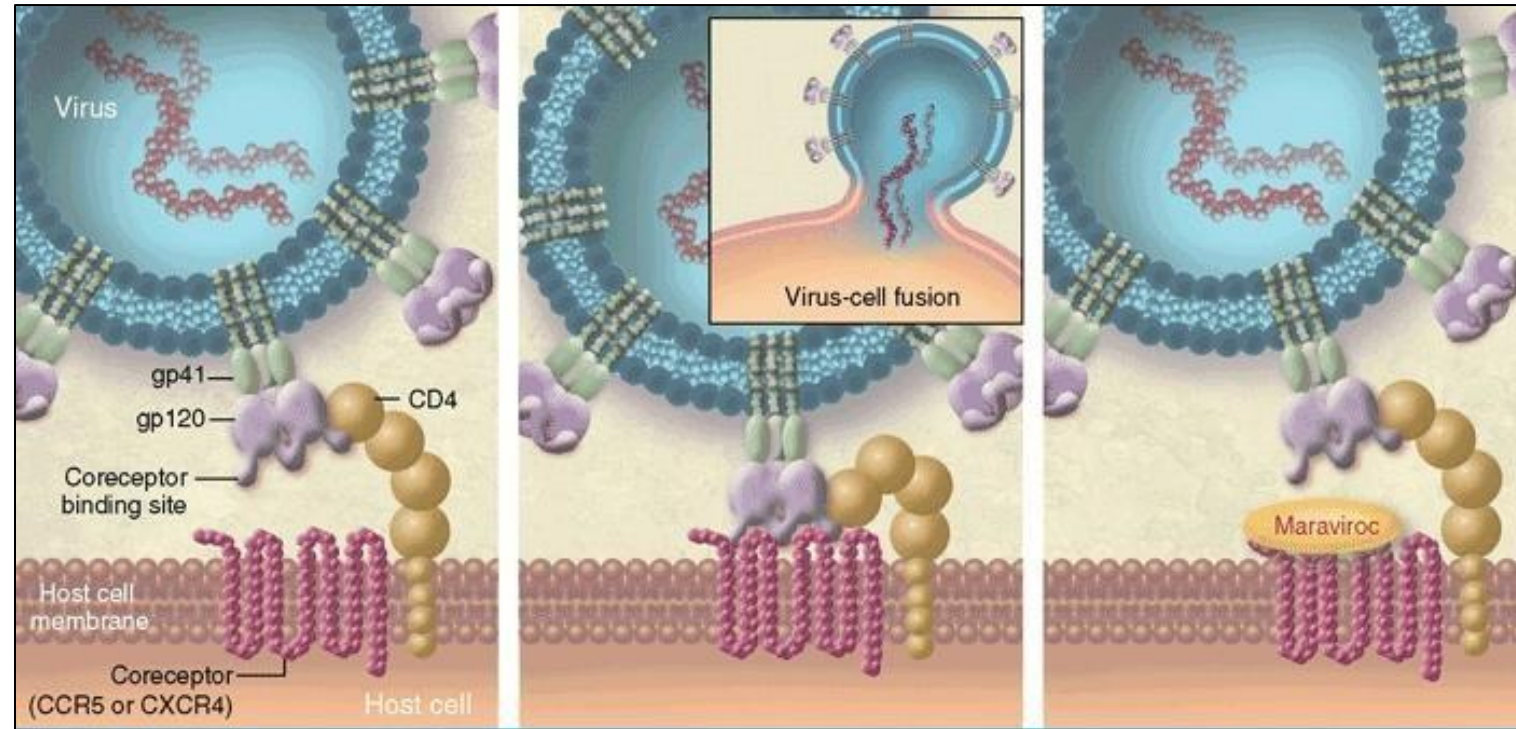
Maraviroc (Selzentry)

MOA - **fusion inhibitor** - binds to the **CCR5 receptor** of the CD4 T-cell – one of two chemokine receptors necessary for entrance of HIV into CD4+ cells.

Use – oral; **ONLY** used in patients with **CCR5-tropic HIV infection** in whom other treatment has not been effective

- **NOT** for CXCR4 or mixed tropism – testing should be performed

Adverse effects - generally safe and well tolerated; rash common



Fostemsavir (Rukobia)

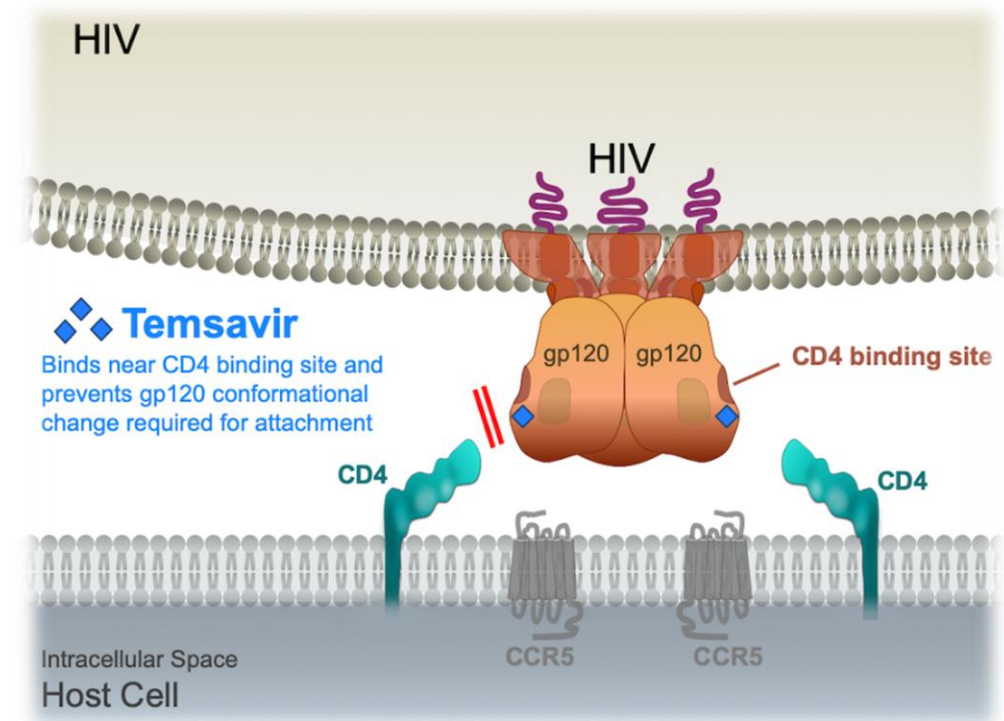
- Oral prodrug
- Approved 2020

MOA - **Binds to gp120 subunit**, preventing interaction with CD4 receptor and entry of virus

Use - treatment-resistant patients

Adverse effects

- May increase hepatic enzymes, bilirubin, cholesterol and triglycerides
- Nausea and diarrhea possible



Ibalizumab (Trogarzo)

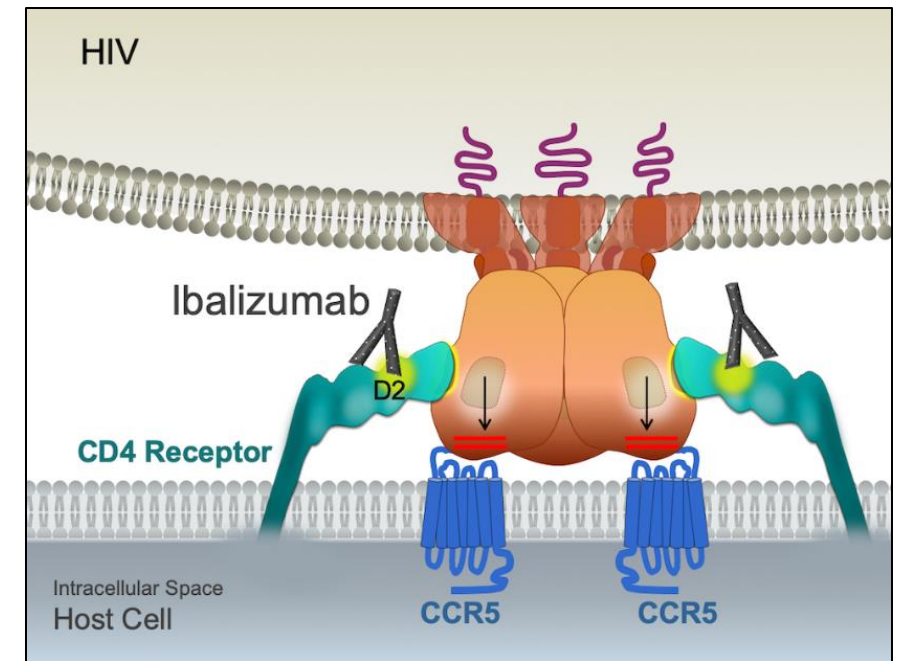
MOA - Binds to the **CD4 receptor** to block viral entry

- prevents conformational changes in CD4 and gp120 that would allow virus to enter

Use - Effective against **both CXCR4- and CCR5-tropic strains**

- For patients when other treatments begin to fail; combined with other treatments
- Long acting, IV every two weeks

Adverse effects - diarrhea, dizziness, nausea, and rash



reverse transcriptase inhibitors

NRTIs – abacavir, didanosine, emtricitabine, lamivudine, tenofovir, zalcitabine, zidovudine
NNRTIs – atevirdine, delavirdine, efavirenz, nevirapine, pyridinones

